D06 · Logistics Technology & Innovation

04 · Digital twins & simulation in logistics network planning

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

A logistics digital twin is a data-fed model of the network — nodes, flows, capacities, inventories and rules — used to simulate changes and disruptions before they happen in the physical operation.

2 WHY IT MATTERS

Simulation lets a team test a DC closure, a carrier failure or a new policy without paying for the mistake in reality. **Its usefulness depends on how current and accurate its data is**; a twin that still contains a closed facility gives confident, wrong answers. Research on supply chain twins highlights their value for disruption and resilience analysis **(Ivanov and Dolgui, 2021 — Production Planning and Control)**.

3 HOW IT IS DONE

1. **Start with a painful question.** DC outage, border closure, carrier loss or network redesign — not a generic model.
2. **Connect the data pipelines.** Master data, flows and capacities refreshed automatically, with defined latency.
3. **Validate against history.** Reproduce a known past period before trusting the model on the future.
4. **Embed it in decisions.** Use scenarios in S&OP and network reviews, and keep the twin synchronized with reality.

IN PRACTICE

Invest in data pipelines before exotic simulation features

Garbage-in-garbage-out is especially dangerous when the output looks precise and authoritative.

Start with the highest-pain scenarios

DC outage, key carrier failure and border disruption usually deliver the fastest insight and organizational learning.

Keep the twin synchronized with reality

A twin that still contains a closed facility or an obsolete lane produces misleading recommendations.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Pharmaceutical distributor stress-testing a 72-hour main-DC outage

Scenario Element	Simulated Impact	Prepared Response
Main DC offline 72 h	32% of volume impacted on days 1–2	Pre-defined allocation to secondary nodes
Secondary DC capacity	Covers ~65% of critical demand	Priority rules by clinical criticality
Additional carrier capacity	Confirmed in simulation	Pre-negotiated surge agreements activated
Customer communication	—	Proactive messaging templates ready

Result: When a real (shorter) disruption later occurred, the team executed the pre-tested plan. Customer impact was estimated at less than 25% of the no-plan scenario.

HOW IT IS MEASURED

Digital Twin Data Latency

Maximum age of the operational data reflected in the twin

Benchmark: <4 h for operational twins · <24 h acceptable for tactical planning twins

% of Major Network Decisions Supported by Simulation

(Significant decisions that used twin/scenario analysis / Total significant network decisions) × 100

Benchmark: >80% indicates the twin has become part of the management system

WHAT YOU WOULD USE

Digital Twin & Simulation

Kinaxis Maestro (formerly RapidResponse) / Blue Yonder Luminate — Planning platforms with strong scenario and digital-twin capabilities.

anyLogistix / specialized simulation tools — Used for detailed network and operational simulation studies.

4 SO WHAT

Build the twin around real questions and fresh data. Start with a high-pain scenario, automate the data, validate on history and use it where decisions are actually made.

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